

# Effect of Gyroscopic Couple on Naval Ships

by

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# What is Gyroscope?

## Definition:

A gyroscope is a device for measuring or maintaining orientation, based on the principles of the conservation of angular momentum.

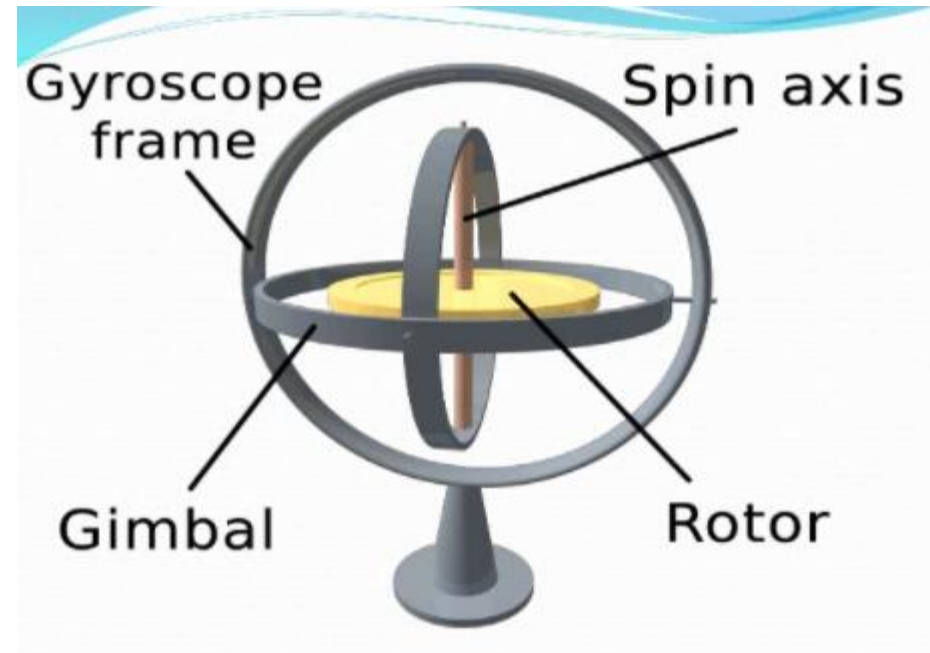


Fig. 1. Gyroscope

- If the axis of spinning or rotating body is given an angular motion about an axis perpendicular to the axis of spin, an angular acceleration acts on the body about the third perpendicular axis.
- The torque required to produce this acceleration is known as the active gyroscopic torque.
- A reactive gyroscopic torque or couple also acts similar to the concept of centripetal and centrifugal forces on a rotating body.
- The effect produced by the reactive gyroscopic couple is known as the gyroscopic effect.
- Thus aeroplane, ships, automobiles, etc, that have rotating parts in the form of wheels or rotors of engines experiences gyroscopic effect while taking turn, such as when axes of spin is subjected to some angular motions.

## Gyroscope contd...

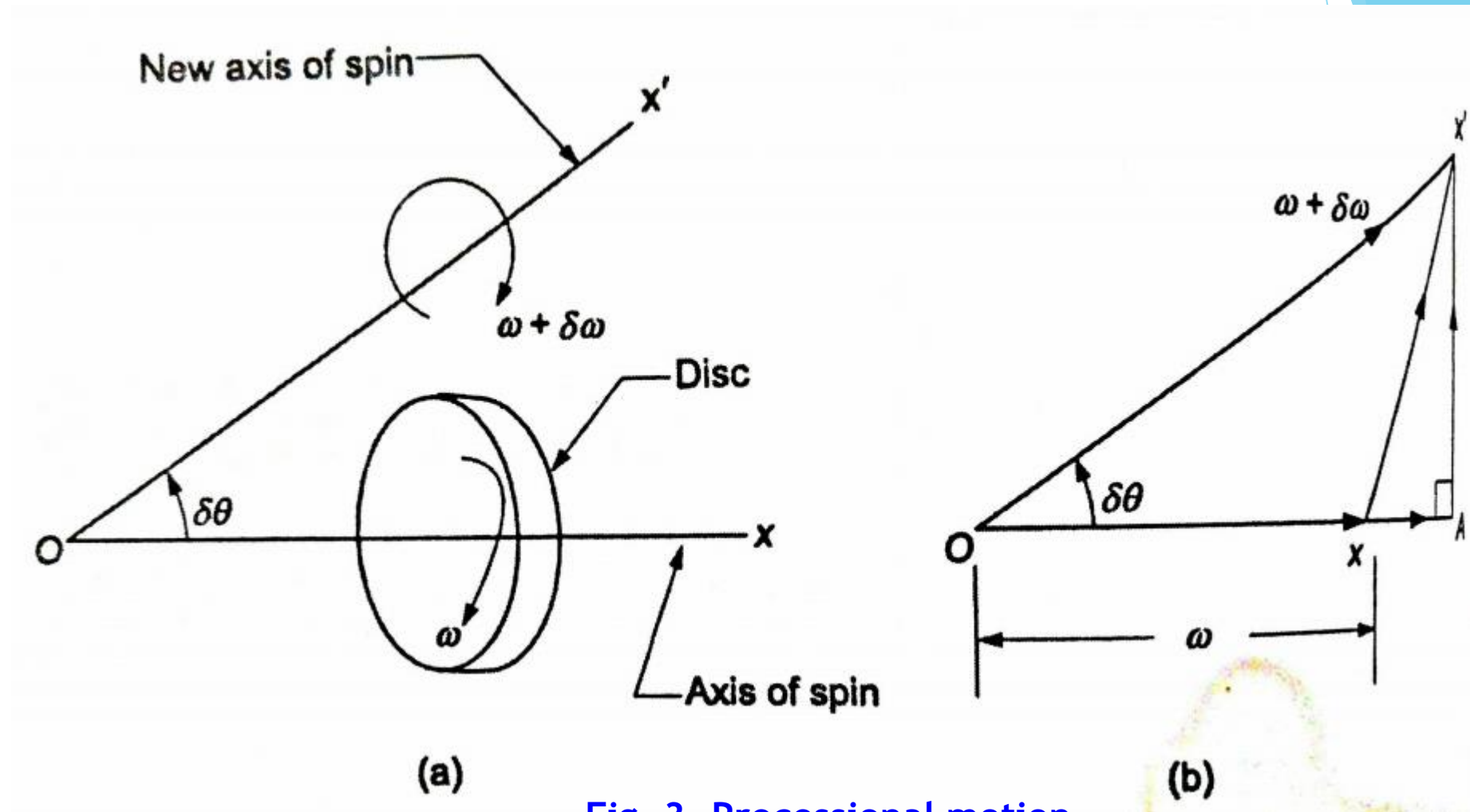


Fig. 3. Precessional motion

## Gyroscope contd...

Angular acceleration along

$$\begin{aligned} OX &= \lim_{\delta t \rightarrow 0} \left[ \frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[ \frac{\omega \cos \delta\theta + \delta\omega \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[ \frac{\omega + \delta\omega - \omega}{\delta t} \right] \quad (\text{Because } \cos \delta\theta \approx 1) \\ &= \lim_{\delta t \rightarrow 0} \frac{\delta\omega}{\delta t} \\ &= \frac{d\omega}{dt} \end{aligned}$$

## Gyroscope contd...

Angular (or gyroscopic) acceleration perpendicular to  $OX$ ,

$$\begin{aligned} &= \lim_{\delta t \rightarrow 0} \left[ \frac{(\omega + \delta\omega) \sin \delta\theta}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \frac{(\omega \cdot \delta\theta + \delta\omega \cdot \delta\theta)}{\delta t} \\ &= \frac{\omega \cdot d\theta}{dt} \quad [\text{Neglecting } \delta\omega \cdot \delta\theta, \text{ being small}] \\ &= \omega \cdot \omega_p \end{aligned}$$

where  $\omega_p$  is the precessional angular speed of the spin axis.  
Total angular acceleration of disc =  $XX'$



## Gyroscope contd...

$$\begin{aligned} &= XA \quad AX' \\ &= \frac{d\omega}{dt} \quad \omega \cdot \frac{d\theta}{dt} \\ &= \frac{d\omega}{dt} \quad \omega \cdot \omega_p \end{aligned}$$

where  $\rightarrow$  represents vector sum.

## Gyroscope contd...

Total angular acceleration along  $XX'$  may be written as:

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

1.  $d\omega/dt$ , represents change in the magnitude of the angular velocity of the rotor.
2.  $\omega.d\theta/dt$ , represents change in the direction of the axis of spin, the direction of  $AX'$  is from  $A$  to  $X'$  in the vector diagram (being a component of  $XX'$ ), the acceleration acts clockwise in the vertical plane  $xz$  (when viewed from front along  $y$ -axis).



## Gyroscope contd...

### Fundamentals of Gyroscopic Motion:

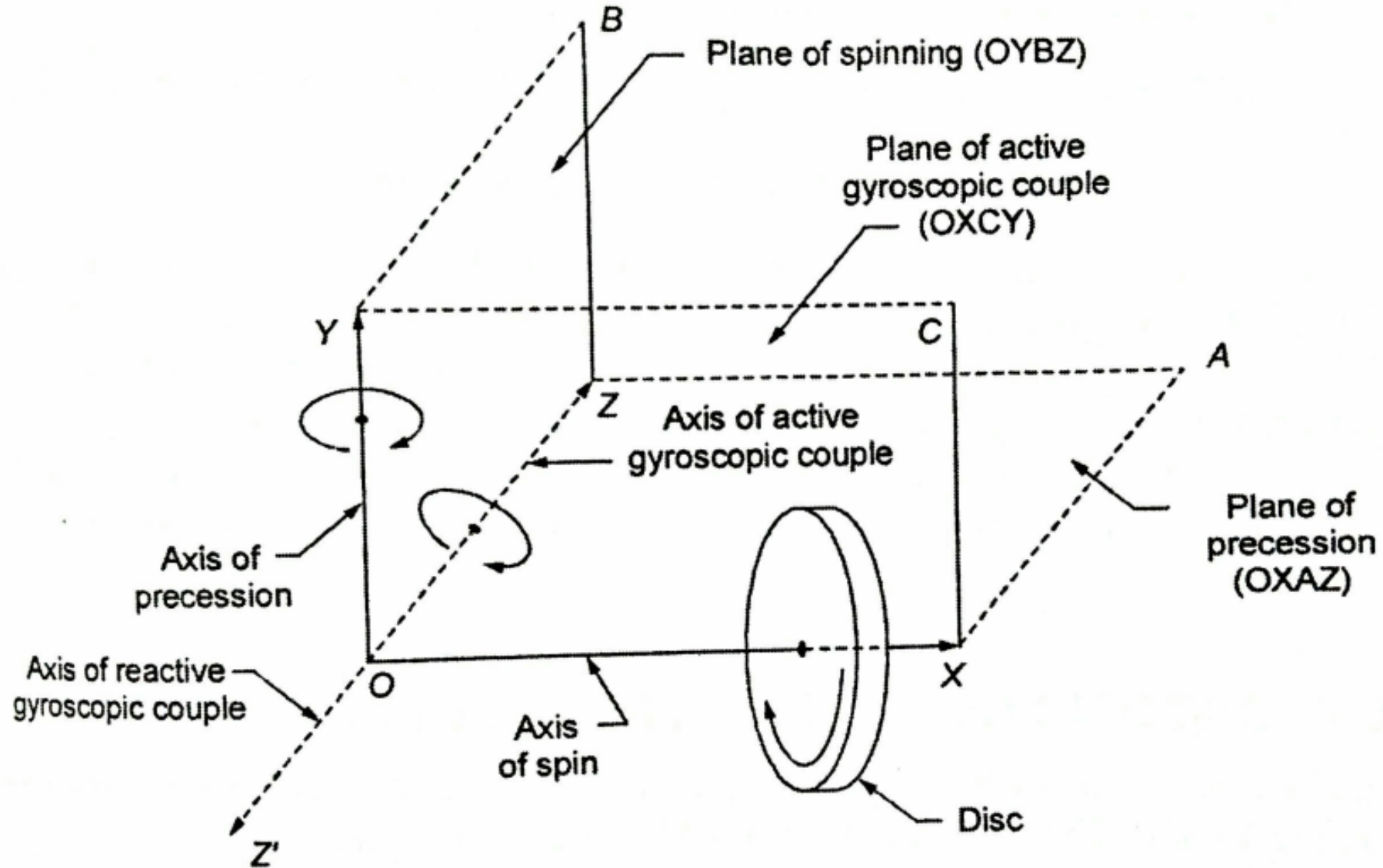
**Gyroscope:** It is a body which spinning about an axis is free to move in other direction under the action of external forces.

**Axis of Spin:** It is axis about which the body revolves.

**Gyroscopic effect:** consider a body spinning about an axis  $OX$ . If couple represented by vector  $OZ$  perpendicular to  $OX$ .

- Then the body tries to precess about an axis  $OY$ , which is perpendicular to both  $OX$  and  $OZ$ .
- This combined effect is called gyroscopic or precessional effect.
- The plane of spin, plane of precession, and plane of gyroscopic couple are mutually perpendicular.

## Gyroscope contd...



**Fig. 4. Gyroscopic effect**

## Gyroscope contd...

**Gyroscopic Couple:**

**Torque or Couple C:**

$$T = I \frac{d\omega}{dt} + T\omega \frac{d\theta}{dt}$$

$$T = I\alpha + T\omega\omega_p$$

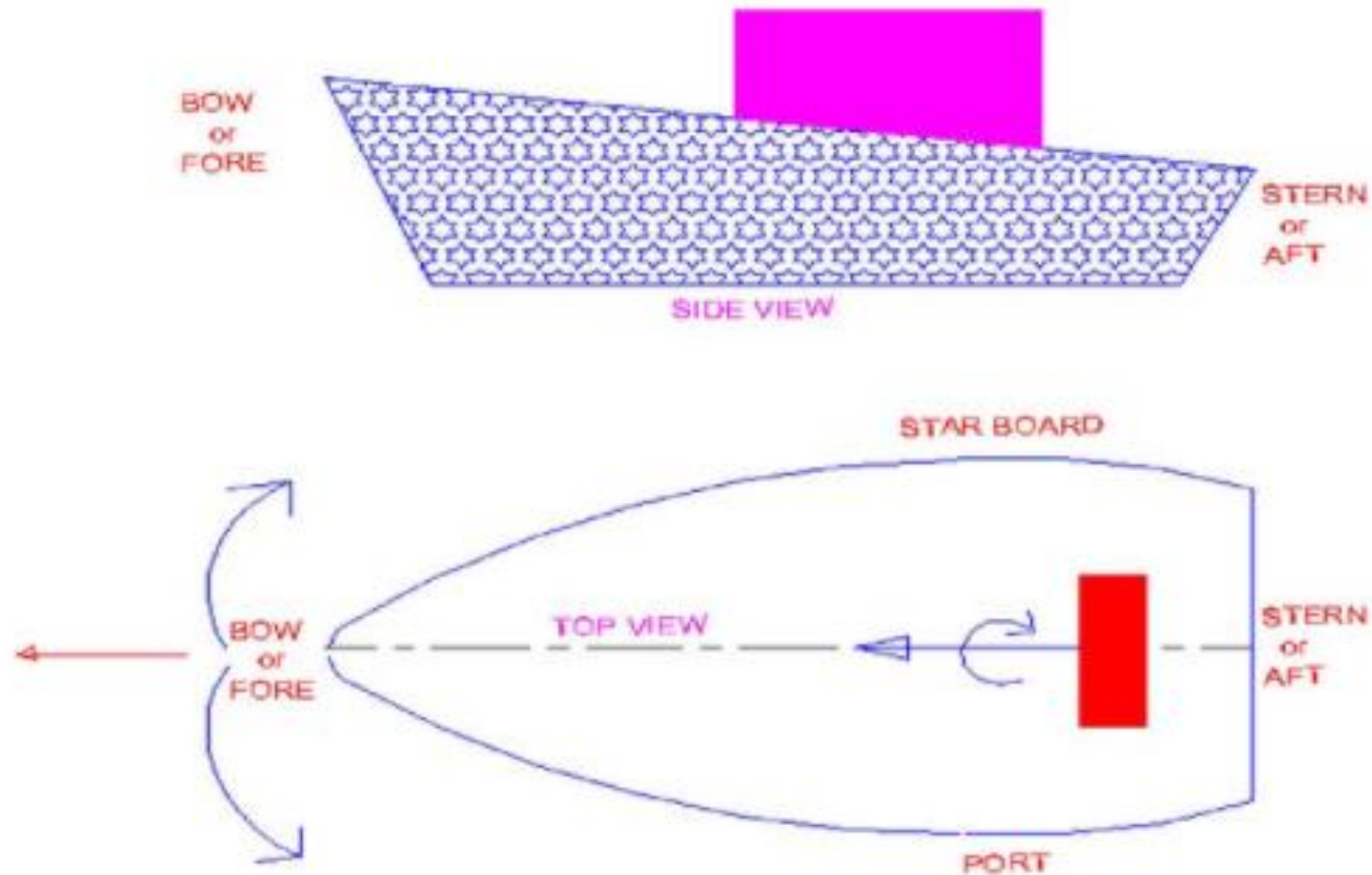
$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

Gyroscopic Couple

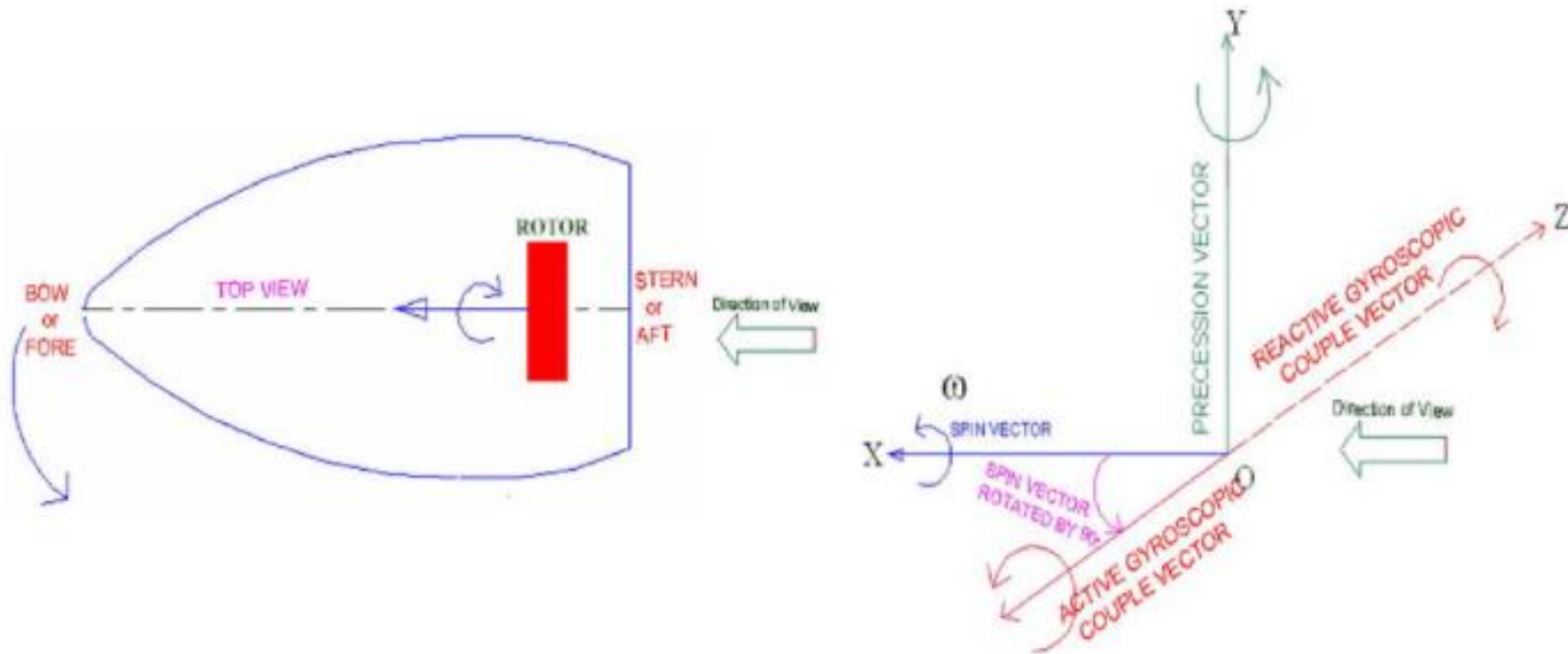
# Gyroscopic Effect on Ship

- Gyroscope is used for stabilization and directional control of a ship sailing in the
- rough sea. A ship, while navigating in the rough sea, may experience the following three
- different types of motion:
- (i) Steering—The turning of ship in a curve while moving forward
- (ii) Pitching—The movement of the ship up and down from horizontal position in a
- vertical plane about transverse axis
- (iii) Rolling—Sideway motion of the ship about longitudinal axis.

- (i) Bow – It is the fore end of ship
- (ii) Stern – It is the rear end of ship
- (iii) Starboard – It is the right hand side of the ship looking in the direction of motion
- (iv) Port – It is the left hand side of the ship looking in the direction of motion

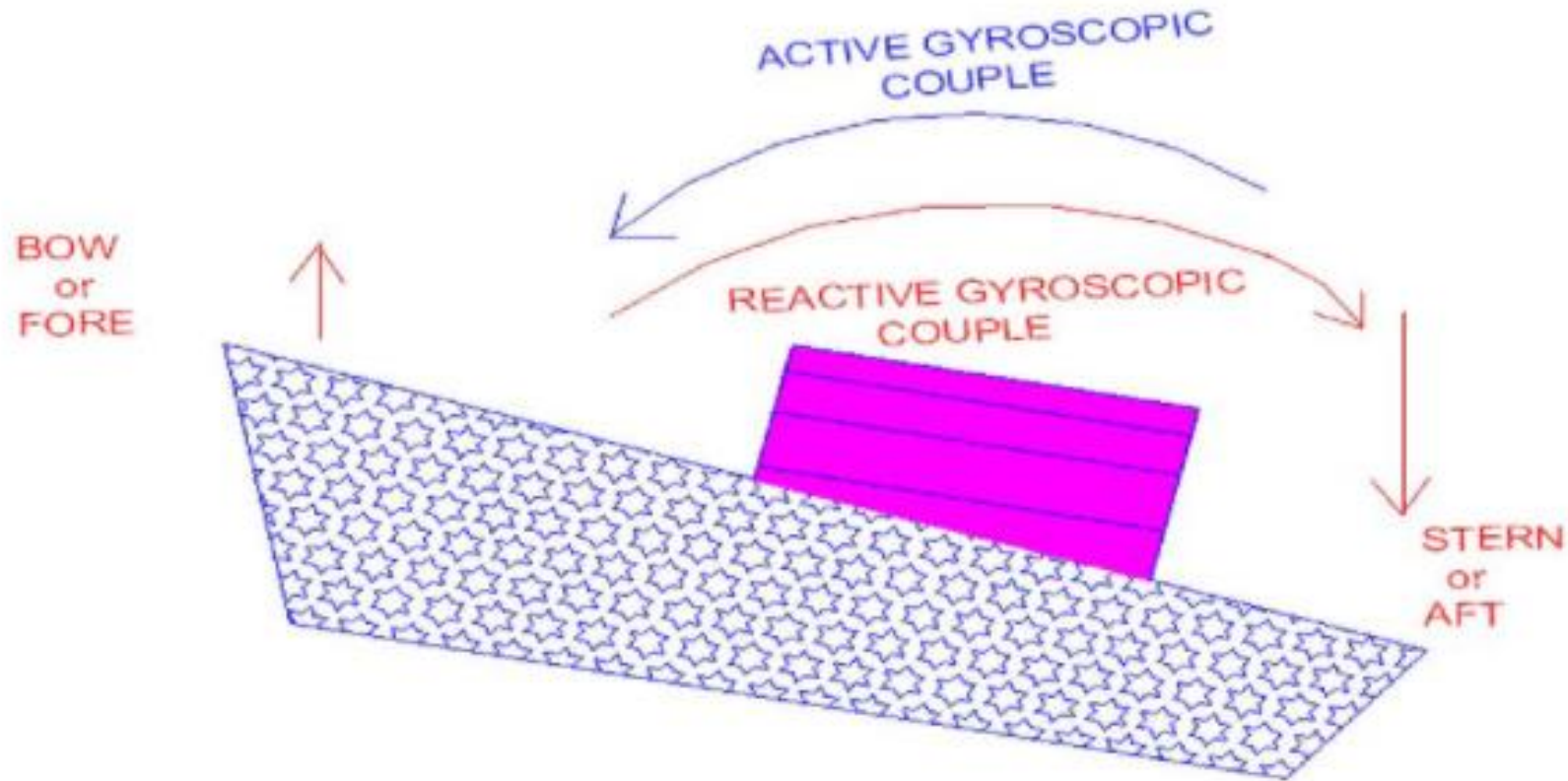


- **(i) Left turn with clockwise rotor**
- When ship takes a left turn and the **rotor rotates in clockwise direction** viewed from stern, the gyroscopic couple act on the ship is analyzed in the following way.



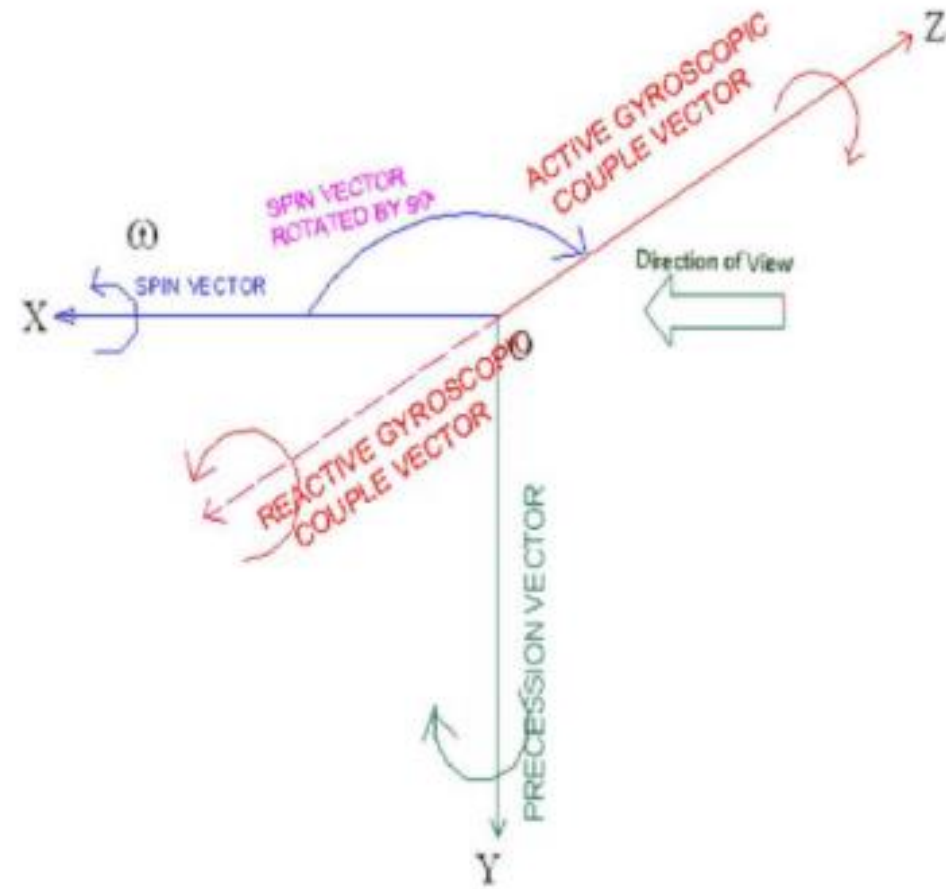
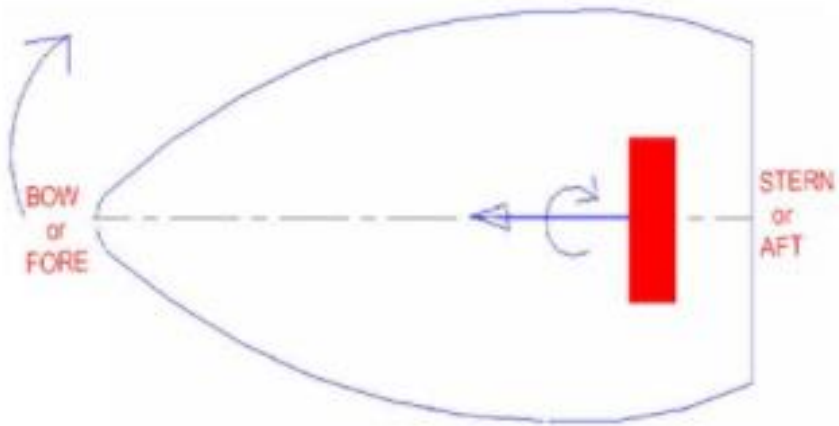


- From the above analysis, the couple acts over the ship between stern and bow. This reaction couple tends to raise the front end (bow) and lower the rear end (stern) of the ship



**(ii) Right turn with clockwise rotor**

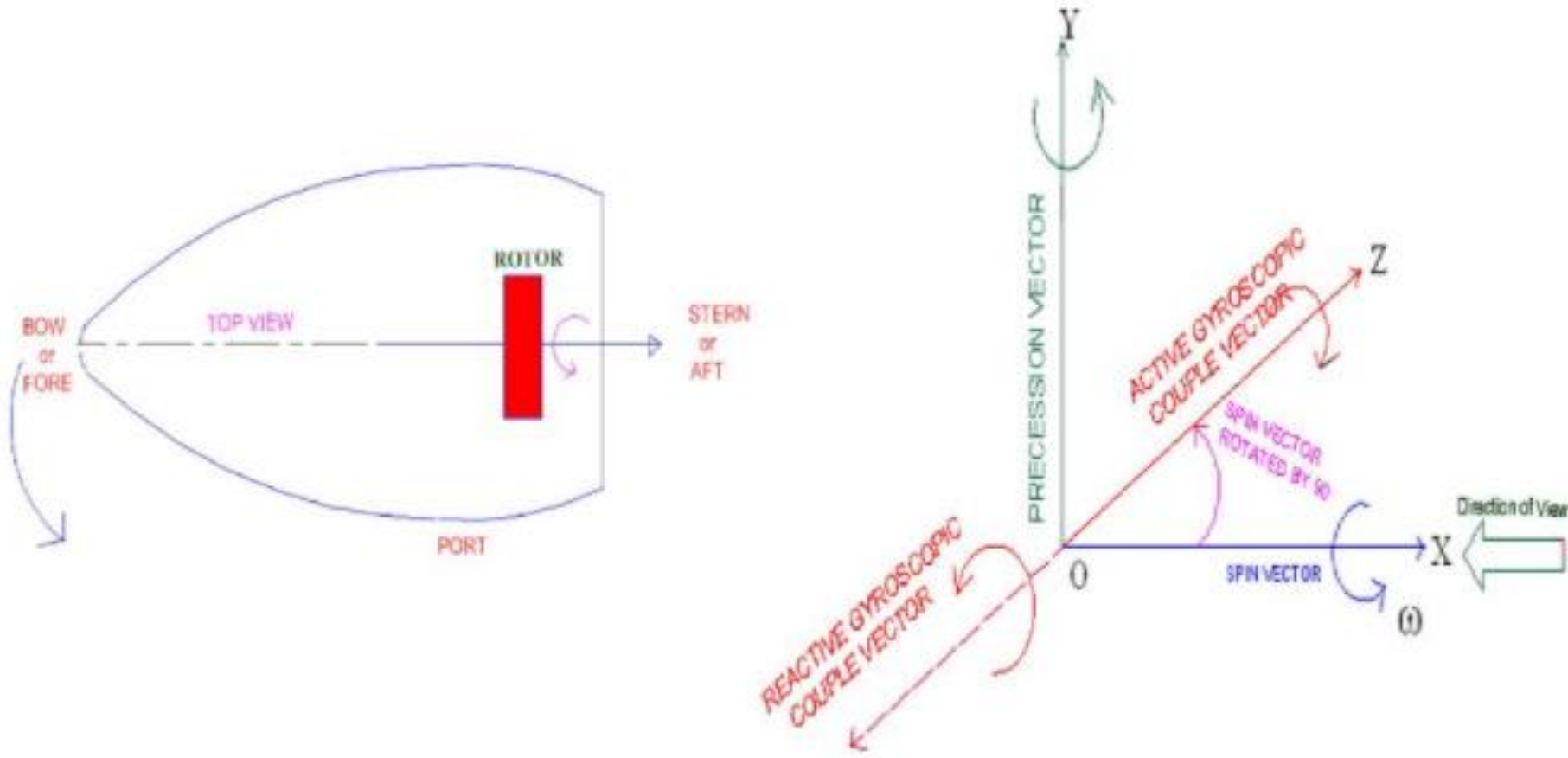
- When ship takes a right turn and the **rotor rotates in clockwise direction** viewed
- from stern



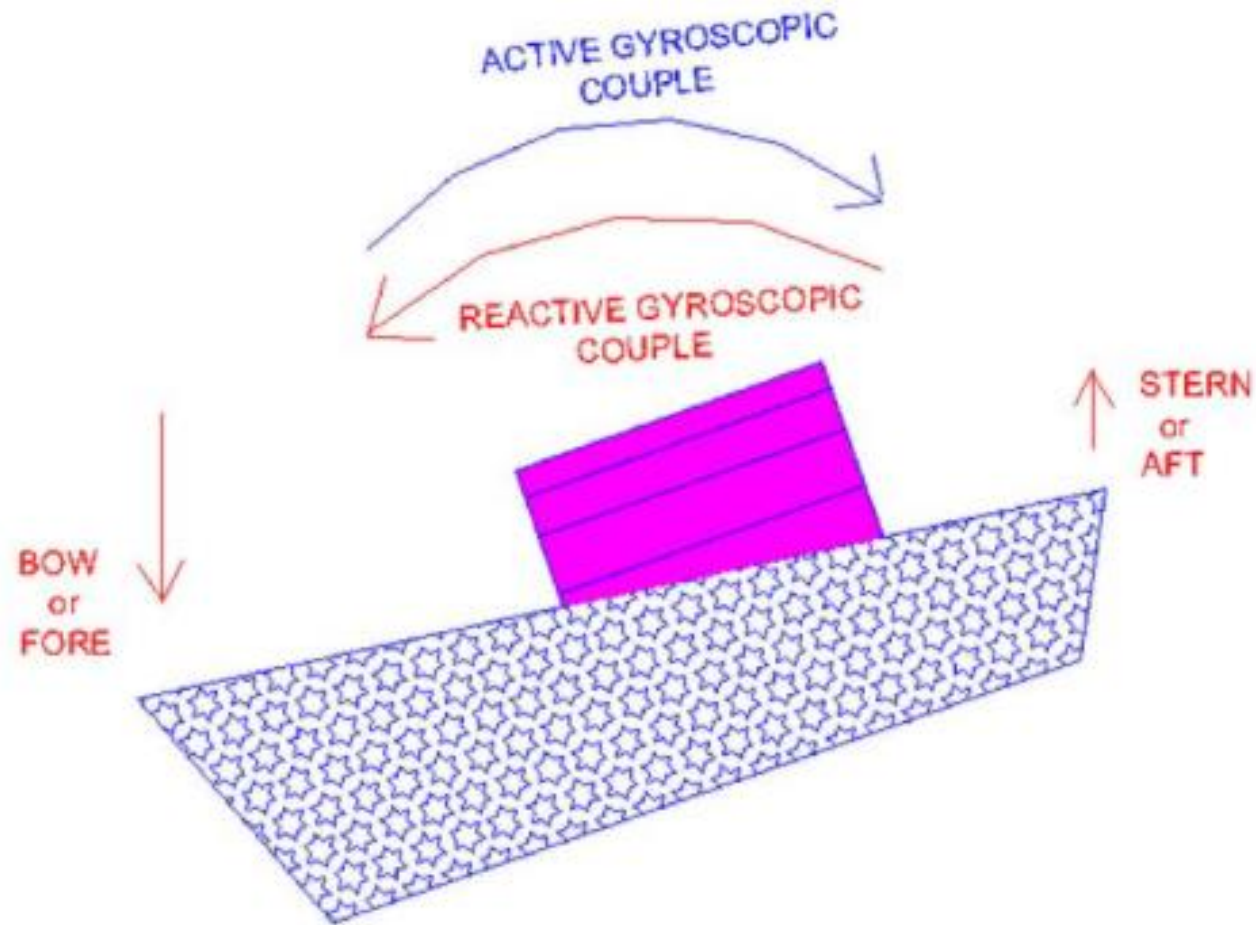


### (iii) Left turn with anticlockwise rotor

- When ship takes a left turn and the **rotor rotates in anticlockwise direction** viewed from stern

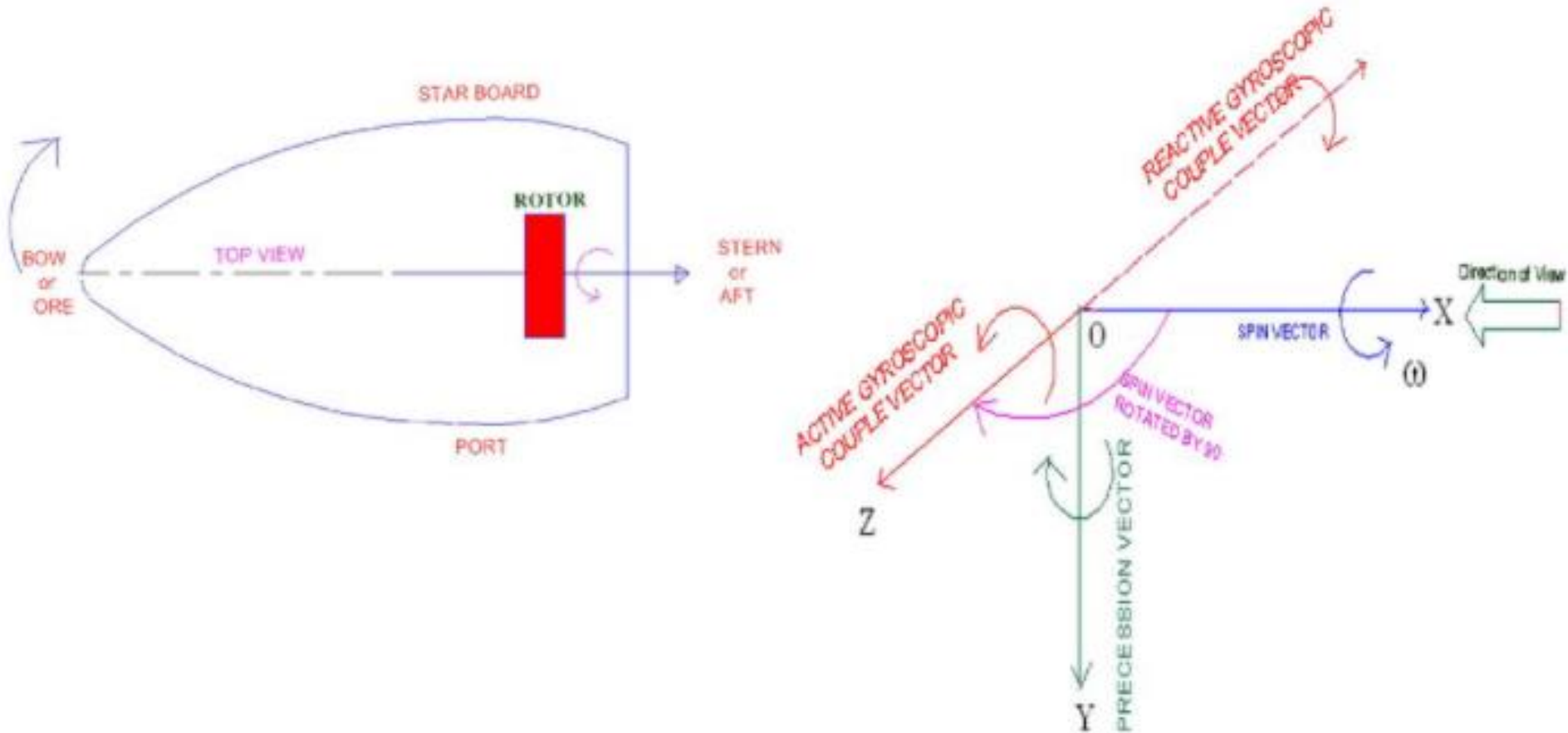


- The couple acts over the ship is between stern and bow. This reaction couple tends to press or dip the front end (bow) and raise the rear end (stern) of the ship.

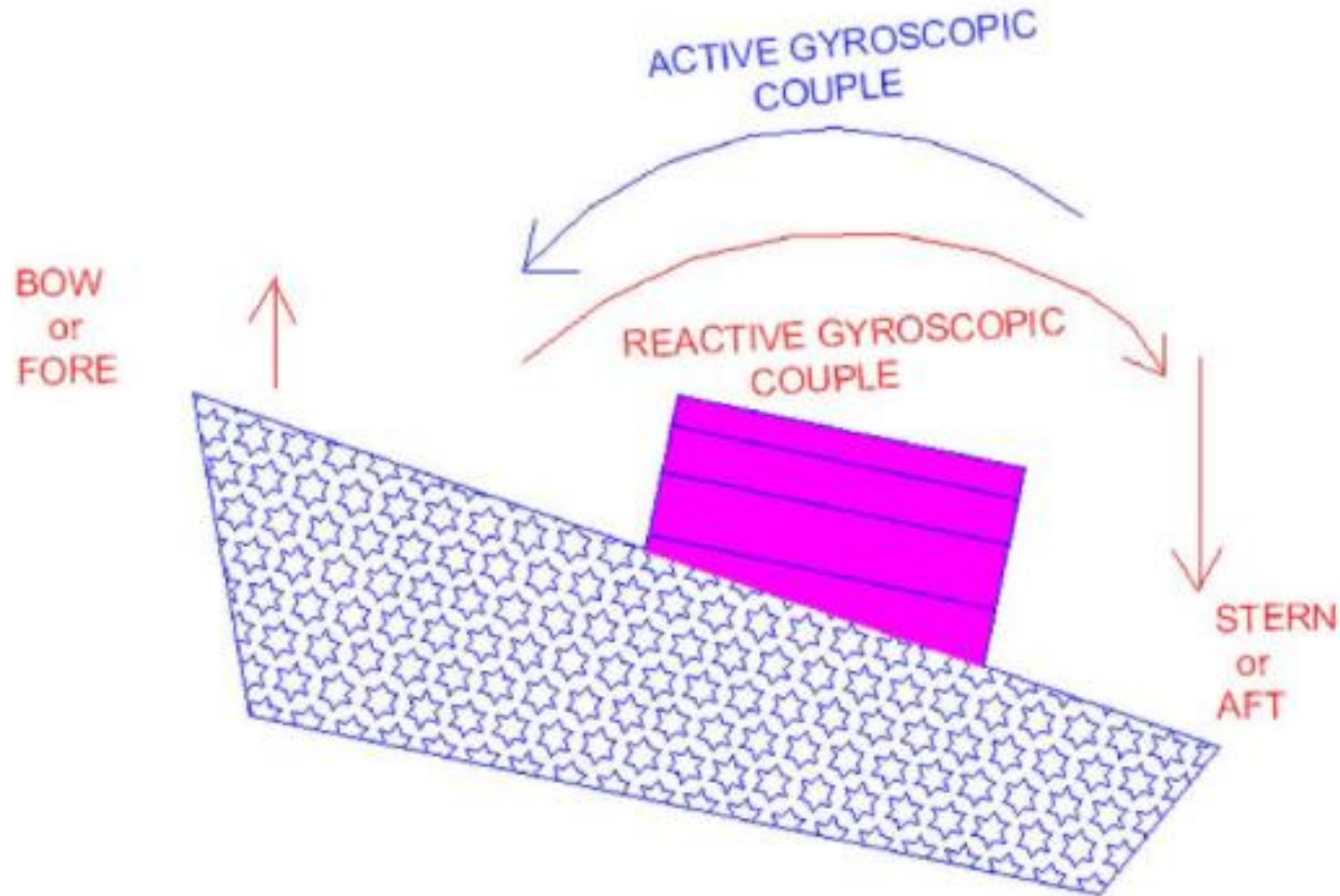


**(iv) Right turn with anticlockwise rotor**

- When ship takes a right turn and the **rotor rotates in anticlockwise direction** viewed from stern



- the gyroscopic couple act on the ship is according to Fig 20. Now, the reaction couple tends to raise the bow of the ship and dip the stern.



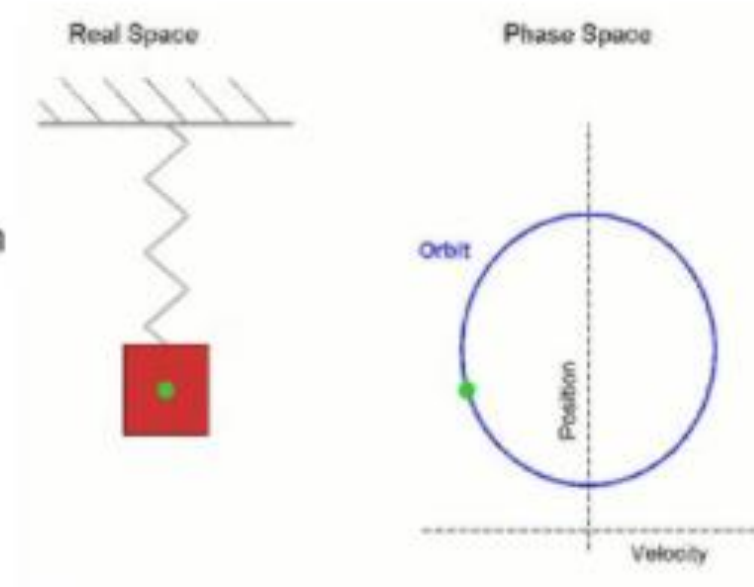
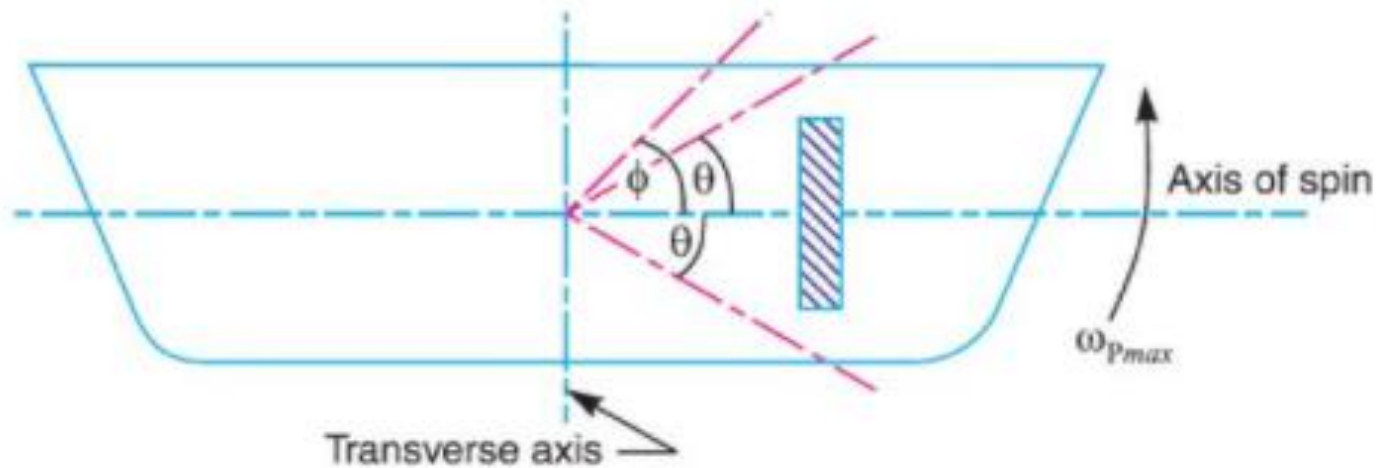
## Gyroscopic effect on Pitching of ship

- The pitching motion of a ship generally occurs due to waves which can be approximated as sine wave. During pitching, the ship moves up and down from horizontal position in vertical plane





- Pitching is the movement of a complete ship up and down in a vertical plane about transverse axis, as shown in Fig.
- In this case, the transverse axis is the axis of precession. The pitching of the ship is assumed to take place with simple harmonic motion i.e. the motion of the axis of spin about transverse axis is simple harmonic.



∴ Angular displacement of the axis of spin from mean position after time  $t$  seconds,

$$\theta = \phi \sin \omega_1 \cdot t$$

$\phi$  = Amplitude of swing *i.e.* maximum angle turned from the mean position in radians, and

$\omega_1$  = Angular velocity of S.H.M.

$$= \frac{2\pi}{\text{Time period of S.H.M. in seconds}} = \frac{2\pi}{t_p} \text{ rad/s}$$

Angular velocity of precession,

$$\omega_p = \frac{d\theta}{dt} = \frac{d}{dt}(\phi \sin \omega_1 \cdot t) = \phi \omega_1 \cos \omega_1 t$$

The angular velocity of precession will be maximum, if  $\cos \omega_1 \cdot t = 1$ .

∴ Maximum angular velocity of precession,

$$\omega_{p_{max}} = \phi \cdot \omega_1 = \phi \times 2\pi / t_p \quad \dots(\text{Substituting } \cos \omega_1 \cdot t = 1)$$

Let

$I$  = Moment of inertia of the rotor in kg-m<sup>2</sup>, and

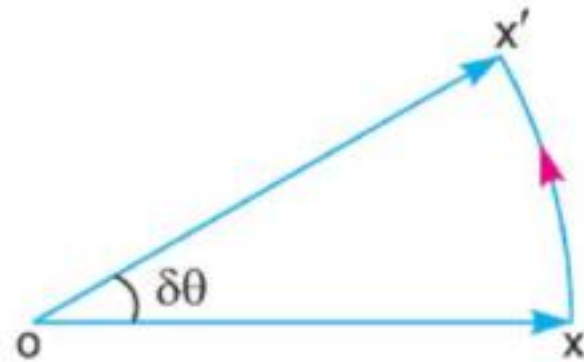
$\omega$  = Angular velocity of the rotor in rad/s.

∴ Maximum gyroscopic couple,

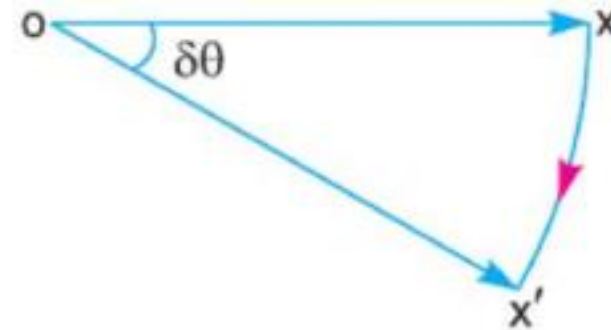
$$C_{max} = I \cdot \omega \cdot \omega_{p_{max}}$$



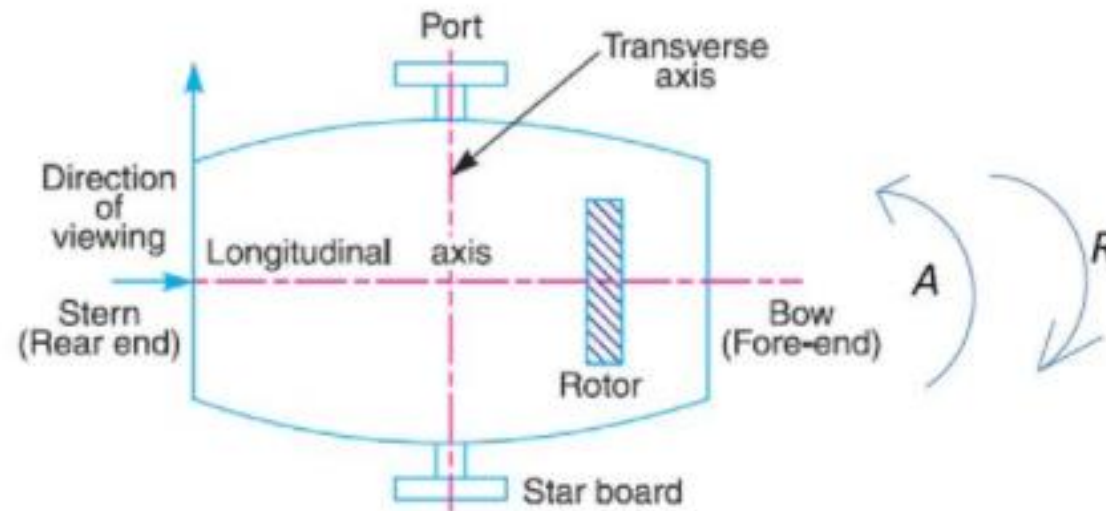
- When the pitching is upward, the effect of the reactive gyroscopic couple, as shown in Fig.(b), will try to move the ship toward star-board.
- On the other hand, if the pitching is downward, the effect of the reactive gyroscopic couple, as shown in Fig.(c), is to turn the ship towards port side.



(b) Pitching upward



(c) Pitching downward



## Notes :

- The effect of the gyroscopic couple is always given on specific position of the axis of spin *i.e.* whether it is pitching downwards or upwards.
- The pitching of a ship produces forces on the bearings which act horizontally and perpendicular to the motion of the ship.
- The maximum gyroscopic couple tends to shear the holding-down bolts.
  
- The angular acceleration during pitching is given by

$$\alpha = \frac{d^2\theta}{dt^2} = -\phi(\omega_1)^2 \sin \omega_1 t \quad \dots \left( \text{Differentiating } \frac{d\theta}{dt} \text{ with respect to } t \right)$$

The angular acceleration is maximum, if  $\sin \omega_1 t = 1$ .

$\therefore$  Maximum angular acceleration during pitching,

$$\alpha_{max} = (\omega_1)^2$$

## Gyroscopic effect on Rolling of ship.

- The axis of the rotor of a ship is mounted along the longitudinal axis of ship and
- therefore, there is **no** precession of this axis. Thus, no effect of gyroscopic couple on the ship
- frame is formed when the ship rolls



**Problem 1.** The turbine rotor of a ship has a mass of 2.2 tonnes and rotates at 1800 rpm clockwise when viewed from the aft (stern). The radius of gyration of the rotor is 320 mm. Determine the gyroscopic couple and its effect when the:

1. Ship turns right at a radius of 250 m with a speed of 25 km/h.
2. Ship pitches with the bow rising at an angular velocity of 0.8 rad/sec.
3. Ship rolls at an angular velocity of 0.1 rad/sec.

Solution: mass  $m = 2200$  kg,  $N = 1800$  rpm,  $R = 250$  m,  $k = 0.32$  m

$$V = 25 \times 1000 / (60 \times 60) = 6.94 \text{ m/sec}$$

$$I = mk^2 = 2200 \times (0.32)^2 = 225.3 \text{ kg.m}^2$$

$$\omega = \frac{2\pi \times 1800}{60} = 188.5 \text{ rad / sec}$$

$$\omega_p = \frac{v}{R} = \frac{6.94}{250} = 0.0278 \text{ rad / sec}$$

(i)  $Couple = I\omega\omega_p = 225.3 \times 188.5 \times 0.0278 = 1180 N - m$

The effect is to lower the bow (fore) and raise the stern (aft) when ship turns right.

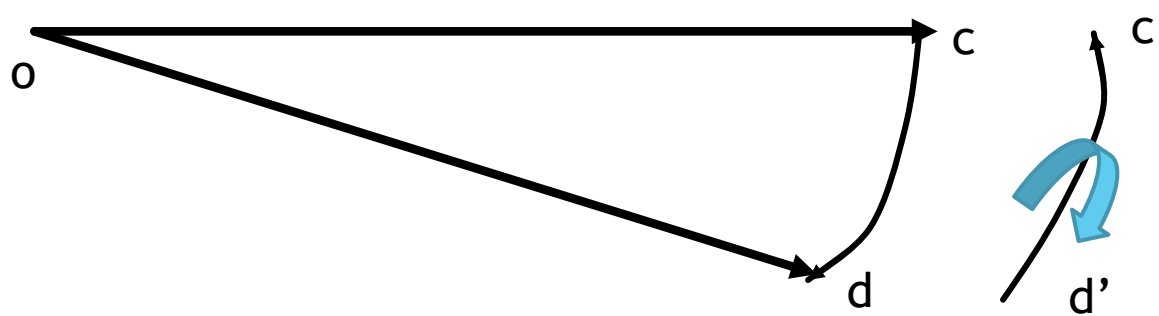


Fig. Right steering

(ii)  $\omega_p = 0.8 rad / sec$

$$Couple = I\omega\omega_p = 225.3 \times 188.5 \times 0.8 = 33972 N - m$$

The effect of the reaction couple when the bow is rising, is to turn the ship towards right or towards starboard.

(iii)  $\omega_p = 0.1 \text{ rad / sec}$

$$\text{Couple} = I\omega\omega_p = 225.3 \times 188.5 \times 0.1 = 4246.5 \text{ N} - \text{m}$$

As the axis of spin is always parallel to the axis of precession for all positions, there is no gyroscopic effect on the ship.



**Problem 2.** The rotor of the turbine of a ship has a mass of 2500 kg and rotates at a speed of 3200 rpm counter clockwise when viewed from the stern. The rotor has radius of gyration of the rotor is 400 mm. Determine the gyroscopic couple and its effect when the:

1. Ship steers to the left in a curve of 80 m radius at a speed of 27900 m/hour.
2. Ship pitches 5 degree above and 5 degree below the normal position and the bow is descending with its maximum velocity – the pitching motion is simple harmonic with a periodic time of 40 seconds.
3. Ship rolls and at the instant, its angular velocity is 0.4 m/sec clockwise when viewed from stern.

Also find the maximum angular acceleration during pitching.

**Solution:** mass  $m = 2500$  kg,  $N = 3200$  rpm,  $k = 0.4$  m

$$V = 27900 / (60 \times 60) = 7.75 \text{ m/sec}$$

$$I = mk^2 = 2500 \times (0.4)^2 = 400 \text{ kg.m}^2$$

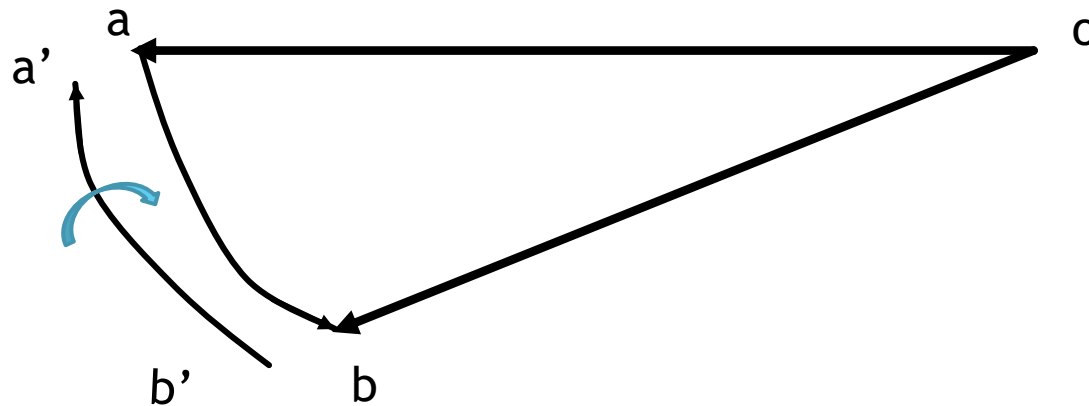
$$\omega = \frac{2\pi \times 3200}{60} = 335 \text{ rad / sec}$$

(i)  $R = 80 \text{ m}$

$$\omega_p = \frac{v}{R} = \frac{775}{80} = 0.097 \text{ rad / sec}$$

$$\text{Couple} = I \omega \omega_p = 400 \times 335 \times 0.097 = 12981 \text{ N} - \text{m}$$

The effect is to lower the bow (fore) and raise the stern (aft) when ship turns left.



$$(ii) \quad \varphi = 5^\circ = 5 \times \frac{\pi}{180} = 0.0873 \text{ rad}$$

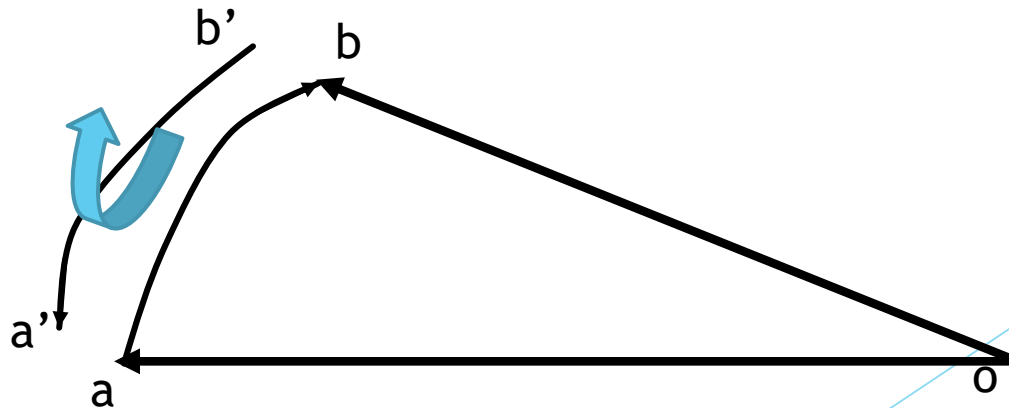
$$T = 40 \text{ second}$$

$$\omega_0 = \frac{2\pi}{40} = 0.157 \text{ rad / sec}$$

$$\omega_p = \varphi \omega_0 = 0.0873 \times 0.157 = 0.0137 \text{ rad / sec}$$

$$\text{Couple} = I \omega \omega_p = 400 \times 335 \times 0.0137 = 1837.5 \text{ N - m}$$

As the bow descends during pitching, the ship would turn towards right or starboard.



(ii)

$$\omega_p = 0.04 \text{ rad / sec}$$

$$\text{Couple} = I \omega \omega_p = 400 \times 335 \times 0.04 = 5360 \text{ N} - \text{m}$$

No gyroscopic effect is there as discussed earlier.

Maximum angular acceleration during pitching:

$$\alpha_{\max} = \varphi \omega_0^2 = 0.0873 \times (0.157)^2 = 0.00215 \text{ rad / sec}^2$$

# Thank You

